

University Lab Consumables Management

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Based on C11 Language | Zero External Dependencies

Linked List Storage | Binary File Persistence | Cross-Platform

2026 Developer Algorithm Programming Challenge - C Language Track Final

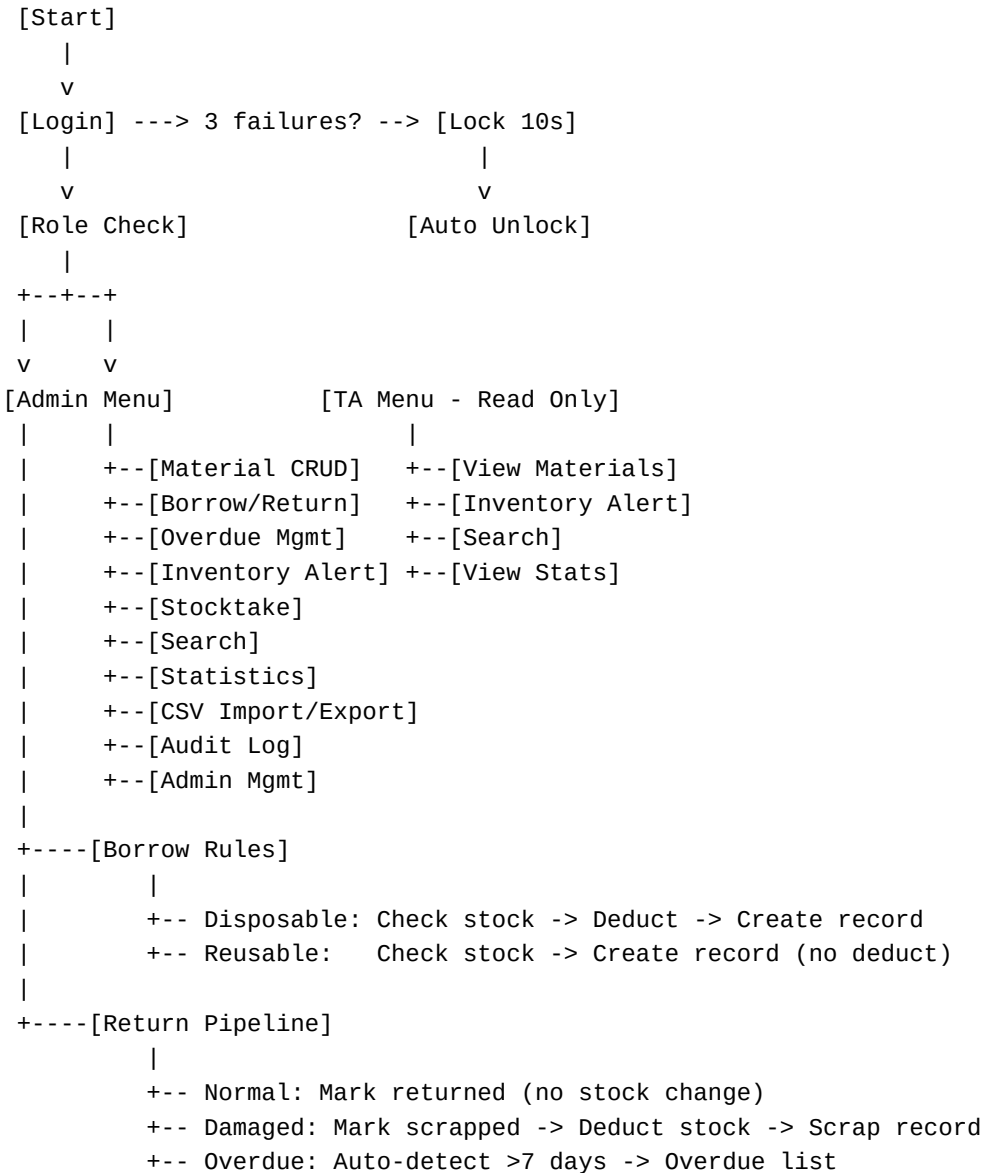
2026-08-04

+-----+

96 Automated Tests | 12 Version Iterations | 5500+ Lines of Code

System Logic Flow Diagram

SYSTEM LOGIC FLOW



DATA PERSISTENCE

All data stored as binary files (*.dat) in data/ directory
Format: [Magic:4B][Version:2B][Count:4B][N x Fixed-Size Struct]
Big-endian byte order for cross-platform compatibility

Module Overview

MODULE ARCHITECTURE

Layer	Modules
Console UI	main.c (menu routing + permission filter)
Business Logic	auth, material, borrow, inventory, search, stats, csv_io, audit (9 modules)
Infrastructure	file_io (binary persistence), ui (screen utilities), platform (cross-platform abstraction)
Data Model	types.h (6 structs + enums + constants)

KEY METRICS

Source files:	13 headers + 13 implementations + 9 test files
Test cases:	96 (all passing, zero compiler warnings)
Data files:	6 binary persistence files in data/
External deps:	0 (C11 standard library only)

Key Data Structures

KEY DATA STRUCTURES (defined in types.h)

Admin	- username, password, role, lock_count, lock_until
Material	- id, name, category(5 types), attr(disposable/reusable), unit_price, total_stock, min_stock, cabinet, purchase_date
BorrowRecord	- record_id, student_id/name, class_name, project_id, material_id, quantity, borrow_time, return_time, status, damage_note, operator_name
ScrapRecord	- scrap_id, material_id/name, scrap_time, reason, quantity, operator_name
StocktakeLog	- log_id, material_id, book_value, actual_value, diff, operator_name, check_time
AuditRecord	- log_id, timestamp, operator_name, action(12 types), target_id, detail

All collections use singly linked lists (no fixed-size arrays).

All string fields use fixed-size char arrays for binary serialization.

1. System Login

Admin login with 3-failure lockout protection

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3. Add Materials

Manual material entry with unique ID validation

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5. Student Borrowing

Differentiated rules: disposable (deduct stock) vs reusable (register only)

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[illegible]

Monthly consumption, class ranking, overdue summary, scrap cost

- [illegible]

[illegible]

Export materials/purchase list/borrow records

[illegible]

Test Coverage Summary

TEST SUMMARY (96 tests, 100% pass rate)

Module	Tests	Coverage

test_file_io	5	Binary file read/write, magic verification
test_auth	15	Login, lockout, role permissions, admin CRUD
test_material	16	Material CRUD, stock ops, scrap, pagination
test_borrow	15	Borrow rules, return, overdue, search
test_inventory	8	Stocktake diff, correction, alert, purchase
test_search	11	Exact match, fuzzy search, multi-filter
test_stats	5	Monthly aggregation, ranking, cost calculation
test_csv_audit	10	CSV import/export, audit logging, filtering
test_integration	11	E2E flow, edge cases, persistence, concurrency

TOTAL	96	

Version History

VERSION HISTORY (12 iterations)

- v0.1 Project skeleton + data structures + file I/O foundation
- v0.2 Login/auth (multi-admin, 3-failure lockout, role-based)
- v0.3 Material management (CRUD, categorization, scrap, pagination)
- v0.4 Student borrowing (differentiated rules, multi-item, receipt)
- v0.5 Return & overdue (batch return, damage-to-scrap pipeline)
- v0.6 Inventory alert & stocktake (purchase list, diff correction)
- v0.7 Search (exact ID, fuzzy name, multi-condition borrow filter)
- v0.8 Statistics (monthly, class ranking, overdue, scrap cost)
- v0.9 Integration tests (E2E, edge cases, persistence, concurrency)
- v0.10 CSV import/export (batch validate, UTF-8 BOM) + audit log (12 types)
- v0.11 Audit moved to module layer + integration test expansion
- v1.2 Final delivery (clang-format, README, 96 tests all passing)

Repository: [git@github.com:Archer11-q/ultc_system.git](https://github.com/Archer11-q/ultc_system.git)